

3. THE POLITICS OF SCOPE: ENDOGENOUS ACTORS, HETEROGENEITY AND INSTITUTIONS

Duncan Snidal

ABSTRACT

The international relations (IR) and common-pool resource (CPR) literatures disagree on the impact of increasing numbers of actors on collective action. They also disagree on the impact of actor heterogeneity. These differences can be understood in terms of the endogeneity of institutions, exemplified here by membership rules. The number and variety of actors need not be an exogenous aspect of a collective-action problem but rather may be determined as part of the institutional rules. Successful CPRs regulate their membership either through exclusion or by enforcing contributions. In some cases this is supported by nesting in external institutions, in other cases enforcement is decentralized and in yet other cases collective action is achieved through locally centralized arrangements. This latter category suggests a much richer range of possible institutional solutions than is found in standard IR conceptions of anarchy or hierarchy. In terms of the various types of actor heterogeneity discussed in the paper, it suggests a need for better understanding of the role of 'community' in supporting collective action at both the local and international levels.

KEY WORDS • collective action • cooperation • heterogeneity • institutions • international regimes

Do large numbers of heterogeneous actors inhibit cooperation? This question is central both to local common-pool resource (CPR) problems and to international relations (IR). Yet no consistent answer emerges across the two settings. The IR conventional wisdom is that large numbers strongly inhibit cooperation, but CPR analysts find no strong independent effect. IR analysts argue that heterogeneity promotes cooperation, but CPR analysts find homogeneity more conducive to cooperation. These puzzles are only partly resolved by recognizing the different forms and thus impact of heterogeneity. The deeper resolution lies in the fact that different institutional settings not only mediate key actor characteristics but determine them. Thus, this paper argues that the number and heterogeneity of actors are not purely exogenous determinants of cooperation but rather are codetermined along with other aspects of institutional scope.

The development of NATO illustrates the codetermination of hetero-

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geneity and scope. The end of World War II did not simply present a few Western states with a well-defined problem that determined the nature of NATO. Instead, considerations of different alternatives led to decisions over membership, issues to include and relations to other international institutions. Choices were made to exclude the former Soviet ally but include Germany and to include some non-European members (US and Canada) but not others (Japan). The issue focus was to be security in Europe and not Asia and economic issues were left to other venues. Indeed, NATO arguably helped create – or at least define – the very collective-action problem of containing the Soviets that it then set out to overcome. Finally, NATO reflected a US decision to pursue security through specialized regional alliances rather than through the UN framework. Once made, these choices provided a relatively fixed framework within which common policies were coordinated for over 40 years.

The post-Cold War era presents a crisis and an opportunity for NATO. Its original mission has greatly diminished in importance while new issues of instability in Eastern Europe and around the globe have gained greater prominence. In considering a recasting of NATO's mission and membership, key issues of institutional design are tightly intertwined with all aspects of the problem. The *scope* of this 'new' institution in terms of membership, issues covered and its new institutional setting will not be strictly determined by the nature of the problem or the participants.¹ Instead, key decisions in the institutional design process will include what states to admit, what issues to handle and how NATO should connect with its broader institutional setting.

This paper examines the impact of the membership dimension of scope (number and nature of actors) on collective action in IR and CPRs. But I argue that scope is itself sometimes endogenous so that membership cannot be taken as a fixed parameter of the problem. Instead, the determination of membership, other institutional rules and the outcome are tightly inter-related. This 'simultaneity' is increased within the institutional design approach of this collection precisely because good design is forward-looking and seeks both to establish effective institutions and to create environments that make institutions more effective.

The paper borrows from lessons on achieving collective action in local

1. Institutional scope refers to the range of inclusion and extent of effective coverage of institutions and rules. In this paper I focus on membership while only implicitly and briefly discussing other aspects of scope, including issues covered, temporal scope and nested relations among various institutions and rules systems. Ostrom's (1990: 90) institutional design principles of 'clearly defined boundaries' and 'nested enterprises' are directly related to scope. Note that scope differs from scale as discussed in papers 1 and 2 in this collection by Young and by Keohane and Ostrom.

CPR problems² and examines their applicability to IR. Comparisons across these different contexts raise questions about the impact of the number and heterogeneity of actors that clarify the more general use of those concepts. The first section of the paper addresses the analytic similarities and possible contextual differences that affect transference from CPR experiences to the IR realm. Even emphasizing the differences between the settings, it is clear that some instructive lessons are available across them.

The second section expands on the methodological point introduced above that institutional scope cannot be considered simply as a fixed environmental factor. The reason is that the choice of scope rules helps determine other key variables including the heterogeneity of actors, which is the key independent variable of this volume. Because our CPR observations are primarily institutional equilibria resulting from protracted but largely unobserved processes, sorting out issues of exogeneity and endogeneity is more difficult than usual. Even where we can observe the sequence of institutional development more closely, the forward-looking nature of institutional design complicates ordinary causal analysis. Thus, extra care is required in specifying the interrelations among the key variables.

The following two sections compare the impact of the number and heterogeneity of participants in the CPR and IR settings. Some of the observed differences across IR and CPR are due to inherent differences in the problems faced, while others are due to different types of heterogeneity. But the greatest difference is due to political-social factors that make membership rules endogenous and alter the incentives and therefore the behavior of actors. The different degrees of success in achieving cooperation between CPRs and IR, or between examples within each area, are then better explained by features of the political setting than by the numbers or heterogeneity of actors. The concluding section summarizes some general implications from this comparison with CPRs for understanding the role of institutions in promoting collective action in IR.

Comparing CPRs and IR

Local CPRs and IR have exploitable similarities because both involve collective-action problems, broadly defined, where independent behavior leads to collectively suboptimal outcomes. However, the two research agendas emphasize different variants within this general category. Local

2. I use Ostrom (1990; see also Ostrom et al., 1994) as an icon for the larger body of local CPR research in several disciplines. Ostrom's work is of special importance because of its theoretical thrust and its ambitious attempts to generalize across cases.

CPR analysis focuses on the provision and appropriation of goods that are not joint in consumption (like private goods) but where exclusion is difficult (like public goods). Standard cases are natural resources, like forests or water, where the quantity available is less than the desired consumption of potential appropriators. This particular problem occurs at the international level as well (e.g. global commons issues) but is not nearly as central in the literature. Much greater attention (probably too much) has been paid to the provision of international public goods that are both joint and non-exclusive.³ More recently, there is an increasing awareness among IR scholars that no single model of collective action deserves pride of place and that different models are appropriate for different problems (Stein, 1982; Snidal, 1985a; Krasner, 1991; Martin, 1992). The obvious corollary is that great care needs to be taken in transferring design principles discovered through analyzing CPRs to any IR problems better characterized as public goods or by other models.⁴

Several other possible differences between local CPRs and IR should be noted. First, local CPRs often *appear* more clearly bounded by physical and technological parameters than do typically messy IR problems. The Law of the Sea negotiations, for example, included not only global fishing matters but issues of navigation and mineral exploitation as well as general consideration of economic development and global equity. For other international issues such as money, nuclear proliferation or migration, the nature and even existence of a collective-action problem remains open to definition. By contrast, while collective action is difficult in the stereotypical CPR problem, it is greatly simplified insofar as the technology of the resource defines the issue, the actors affected by it and their possibilities for action. However, Ostrom (1990) shows that many real CPRs are not so clear-cut and finds the absence of such 'boundary conditions' a prime contributing factor to their failure. Her fishery examples are plagued by outsiders who move into a CPR and frustrate local attempts to manage the resource. Similarly, complex interconnections of underground water sources mean physical boundaries of water basins are not understood and this impedes collective action. On closer analysis, local CPR problems are not inherently more clearly bounded than

3. Public goods is the category of analysis against which Ostrom (1990: 32) is reacting as *partially* inappropriate for CPR problems because they differ on the property of jointness.

4. The other two categories defined by the jointness-exclusion typology are also of interest in international relations. One is that of *private goods* (i.e. exclusive but not joint), which is relevant to many problems such as international trade. Note that the IR literature often transforms this to a public goods problem of providing appropriate trade institutions, a move that sometimes obscures as much as it reveals. The fourth case of *club goods* (joint and exclusive) has been surprisingly underexploited as a way to understand international organizations that provide benefits only to their members.

IR problems and, when they have clearer boundaries, the explanation often lies with political or social factors rather than simply technological ones. Thus, clear boundary definitions are often as much a result as a cause of successful cooperation.

A second substantive difference is that the IR sociopolitical environment *appears* very different from various CPR settings. The convention among IR scholars is to presume an anarchy where security concerns impede cooperation and lawlessness is a major problem. One consequence is a strong emphasis on regimes as providers of the public good of 'order' otherwise absent from international affairs. This difference should not be overstated, however, since CPRs often involve the very livelihoods of the participants. Their security is very much at stake and violence is a very possible outcome.⁵ A more important point is that, even where formally anarchic in the sense of the absence of a strong central government, local CPRs are often embedded in well-articulated societies ranging from peasant villages to highly institutionalized modern democracies. Successful solutions are usually nested in broader social-political relations that support local agreements. But comparison to CPR arrangements contains an instructive lesson that external support need not entail the strong central enforcement implicit in the standard IR anarchy-hierarchy dichotomy (Waltz, 1979).

A third difference is between successful CPR institutions that have evolved and prospective IR institutions that need to be designed. Although many local CPRs developed over extended periods, we have no good record of their evolution, but only a snapshot of the final institutional equilibrium. The final design principles may tell us little about how they were achieved; successful CPR arrangements might even be unintended consequences unattainable by rational design.⁶ Moreover, evolutionary design principles

5. See the description of violence among Valença fishers (Cordell, 1972 in Schlager, 1994), which might reassure the most hard-core realists, although they might find its gradual elimination (not through power-balancing) troublesome. See also Ostrom's examples of violence among Mawelle fishers (1990: 152) and on the Spanish huertas (1990: 77) where participants were orderly in terms of their collective CPR rules, even though their larger society was marked by significant potential for violence. I thank Elinor Ostrom for correcting my tendency in an earlier draft to exaggerate the extent to which the IR setting was distinguished by greater insecurity and violence.

6. Bardhan (1989) cites a compelling example from Langlois (1985). A regulatory institution created by populists to limit monopoly power may then be captured by the capitalist interests it was intended to regulate. The resulting agency is maintained because it serves the interests of large industry even though the principles behind its operation are exactly contrary to the principles embodied in its design. Ironically, such a system of captured agencies probably could never have been established through rational design. This illustrates the possibility of rather sharp divergence between evolutionary and rational design principles. I thank Michael McGinnis and Robert Keohane for a stimulating conversation regarding this issue.

may be inappropriate for other settings. A crude but sufficient example is found with early aspirants to flight who looked towards the evolutionary principles embodied in birds. Wings were a good idea, but flapping them was not a good idea. Alternative design principles for flight, embodied in helicopters and rockets, require no wings and have no direct counterpart in the animal world. This does not mean, of course, that evolutionary examples are not useful – only that they must be used with caution. A design principle that occurs in a successful CPR setting cannot be taken as necessary or sufficient for other problems.

Thus, differences between CPRs and IR are great enough to provoke skepticism in transferring conclusions from one realm to the other, but not so great as to make that infeasible. Indeed, variation within each category on these key factors is probably as great as the variation across categories. The greatest difference is not the technological nature of the underlying issues, or even the political–social environment that surrounds them. Instead, it lies in their combination whereby politics redefines problems to facilitate their solutions. Here lies the strongest parallel between the local CPR literature and the ‘regime’ literature in IR. Each explores how self-interested actors regulate their behavior in mutually beneficial ways through rules and institutions that sit between the unregulated ‘self-help’ behavior of anarchy on the one hand and the strong central regulation of the state or world government on the other. The conceit of the institutional design approach is that principles that work in one arena will work well in the other with appropriate tailoring to particular circumstances. Because our observations are final institutional equilibria and given the forward-looking nature of institutional design, however, it is difficult to sort out issues of cause and effect. This requires us to pay special attention to issues of endogeneity addressed in the next section.

Endogeneity, Exogeneity and Institutional Design

Specifying what is exogenous versus what is endogenous, or independent versus dependent variables, is a difficult first step in theoretically informed empirical analysis. Normal issues of whether our theoretical knowledge is sufficient for this task are further complicated by key properties of institutional analysis. One is that we often observe only the end result of institutional change (e.g. the evolved resolution of a CPR problem) rather than the sequence of steps through which the final institutions developed. This makes it hard to sort out causal patterns among variables that we observe only in

the final institutional equilibrium.⁷ Second, because actors are forward-looking in considering institutional design, current choices will anticipate future conditions that themselves are in part a product of current choices. This further complicates the attempt to sort out cause and effect. In the extreme situation, 'everything is endogenous', which is a recipe for intellectual chaos since it is impossible to find firm ground on which to draw meaningful conclusions about causal processes.⁸

Proper specification of an empirical argument also requires clear distinctions among variables and parameters. Variables are factors whose values fluctuate either for reasons outside the model (exogenous) or for reasons specified inside the model (endogenous). Questions of exogeneity or endogeneity are typically fought out over variables since their status is seen as open to debate. By contrast, parameter values are usually fixed in the model and therefore exogenous. This specification is a claim that the fundamental processes governing behavior are constant over the time frame covered by the model. Note that the fourth category in this fourfold categorization – endogenous parameters – is not typically used.⁹ Below, I argue that this missing category is useful for thinking about the relations among institutions, performance in collective problems and change.

Institutions are usually treated as exogenous parameters that provide the

7. For related discussions on institutions as equilibria see Schotter (1981) and two terrific papers by Calvert (1992a, b). Calvert builds on Schotter to argue that institutional analysis should focus on equilibria as providing simultaneous accounts both of the institution and of behavior within the institution. The current discussion accepts that basic point, but parts with Calvert (1992b: 13, 61; 1992a: 29) insofar as he suggests that the equilibrium analysis 'draws a sharp line between givens . . . and the results' or implicitly defines institutions only endogenously in terms of the equilibrium to the neglect of other aspects of the institution defined exogenously in the rules of the underlying game. Calvert (1992a: nn. 4, 6, 17) recognizes the practical advantages of treating some institutional features as fixed.

8. The econometrics literature is instructive for these issues. On the first point see the controversy surrounding Wold (1960) on the seeming simultaneity in data induced when units of time (i.e. the distance between institutional equilibria that constitute our observation points) are lengthy even if the causal relationships among variables are clear (recursive) on a finer time line. More generally, the identification problem (Fisher, 1966) concerns the impossibility of making sound inferences without a clear set of exogenous variables, supplemented by other theoretical knowledge (e.g. other linear restrictions) about their interrelations. The difficulty increases whenever forward-looking 'rational expectations' make model specification in terms of exogenous and endogenous variables treacherous at best or even impossible (Sims, 1980).

9. In the econometrics literature this corresponds to the case of time-varying coefficients (Jackson, 1992; Judge et al., 1985: chs 13, 19) although there are further complications of simultaneity here.

fixed environment in which social and political processes take place. This is true both in the behavioral tradition and in game-theoretic specification of institutions through the rules of the game.¹⁰ Such a conception accords with the standard normative presumption that the value of institutions lies in providing a stable background against which actors can make intelligent choices. Thus, significant invariance over time is an essential aspect of what we mean by an institution. In the longer term, of course, institutions are affected by the behavior that goes on within them. But this effect is typically slow and cumulative, whether in the development of legal precedent or the gradual evolution of a social custom. Under these circumstances, it is analytically appropriate to treat the institution as parametric and exogenous to the situation.

Institutional change and especially institutional design require us to consider the (partial) endogeneity of institutions. The very notion of institutional design is one of reshaping boundaries between what is endogenous and what is exogenous. Formal institutions create rules to govern the behavior of actors and thereby limit or expand the scope of their endogenous choices. The emergence of informal institutions, such as norms, affects individual behavior by altering the collective belief system that constitutes an important part of each individual's fixed external choice environment.¹¹ In either case, some institutional factor changes, becomes parametric and then constrains or channels future individual behavior.

The normal distinction between exogenous and endogenous variables is thus awkward for institutional analysis. Institutions cannot be considered strictly exogenous since we are interested in how they are created and changed by actors seeking to improve their circumstances. Nor are institutions purely endogenous in the same sense as other choice variables (including how actors behave *given* an institutional setting): institutions matter because they circumscribe and constrain actor behavior, are at least sticky and are sometimes much more rigid than that. For these reasons, institutions are usefully thought of as *endogenous parameters* where the former term indicates that they are susceptible to change and the latter term that they provide a relatively fixed setting for other endogenous choices or

10. Beliefs are the area where game theory has moved closest to endogenizing institutions in the form of updating through the model. However, the beliefs studied tend to concern specific facts of the situation (e.g. is my opponent tough or not?) rather than more general facts about expected behaviors within a multilateral setting. Thus, there has been less progress in building 'focal points' or 'conventions' into the models.

11. 'Fixed', like exogenous, is relative to the time frame. For example, beliefs may be frequently updated and modified and so evolve continuously rather than in discrete jumps. In a sufficiently short time frame, however, beliefs may be sufficiently stable as to be usefully treated as fixed.

variables.¹² The tension between the terms reflects the fact that while institutions are a consequence of human agency and history, they are valuable precisely because they lie outside the domain of short-term choice and thereby constrain it. The important exceptions are those momentous occasions that correspond to constitutional choices and crises. Otherwise, institutions change either slowly through gradual accretion of experience and precedent whose effect is not contemporaneous or else in the guise of small institutional changes that adapt the institution better to immediate circumstances, but do not themselves reflect a fundamental change in the institution.

The status of institutions as usually exogenous, but sometimes endogenous, has been recognized in both the CPR and the IR literatures. Ostrom's distinction between operational, collective-choice and constitutional-choice rules provides an ordering of rules that are progressively more difficult to change and therefore increasingly likely to be exogenous.¹³ Stephen Krasner's (1982) discussion of the difference between changes within regimes and changes of regimes has similar implications for different levels of exogeneity. In any institutional design problem, some rules will be seen as possible targets for change – hence endogenous parameters – and others will be seen as strictly exogenous parameters. Even when we speak of fairly grand 'changes of regimes', however, there must be some defining environment or else, in Ostrom's words, 'the structure of the problem would unravel'. Thus, even changes of regime are almost certainly changes within some larger regime.

A further and significant complication is that forward-looking institutional design implies a cascading endogeneity that can affect other seemingly exogenous variables. Because institutions define environments for future choice and since the choice of design principles depends on their anticipated consequences, good institutional design should select future exogenous variables in salutary ways. Rather than explaining the success of

12. I thank Andrew Dunne for suggesting the term 'endogenous parameters' to describe institutional stability and change. To complete the fourfold typology, other institutions and institutional rules will constitute constant and 'exogenous parameters' in which endogenous institutional change is nested. There also will be 'exogenous variables' whose changes will affect behavior within both evolving and constant institutions. Finally, the key 'endogenous variables' of interest in these analyses will be the behavior of actors under the constraints and opportunities provided them.

13. See Kiser and Ostrom (1982) and Ostrom (1990: 50–5). The distinction here is implicit in the general problem of categorizing variables as exogenous or endogenous. In any dynamic framework, the determination depends on the time period chosen. In the short run, most factors are exogenous, whereas in the very long run almost everything is endogenous. Institutional analysis is complicated because many of its important questions lie in the intermediate territory where there is no easy delimitation of endogeneity and exogeneity.

the institution, these 'exogenous' variables are a consequence of its design. In terms of the focus of the papers in this collection, the number and heterogeneity of actors may themselves be a product of prior institutional choice rather than independent factors. Thus the possibility of institutional design not only makes institutions endogenous: it raises general questions about whether other variables are exogenous or have been affected by the institutional design process.¹⁴

The implications for institutional analysis are twofold. First, it may be difficult to sort out exogeneity, endogeneity and hence causality – especially where design as opposed to evolution is the underlying process. We should look instead for combinations of design principles, circumstances and (favorable) outcomes that are compatible with one another in constituting an equilibrium. Determination of causality among these requires more detailed knowledge of the underlying process, whether it is one of design or of evolution, than may be available. Without that knowledge, we cannot be sure whether an institutional principle is a necessary consequence of a particular environment or if the environment is a consequence of the rule. Second, and for this reason, in discussing long-term institutional change it is important to reconsider the exogeneity of the environment of the institution. For example, the level or cost of exclusion is typically taken as a defining characteristic of CPR or public good problems. But since exclusion itself is a function not only of technical properties of the good but also of legal-political ones, it may be partially endogenous over the longer term. Thus, in understanding the determinants of institutional scope we must also consider the extent to which those determinants are themselves a product of institutional scope.

14. Exogenous parameters other than institutions also can be made endogenous in some circumstances. Even physical constraints are endogenous under conditions of technological change. As Ostrom et al. (1994) point out, for example, we cannot repeal the law of gravity but we can make water run uphill. Similar possibilities of change apply to other exogenous parameters (perhaps including the nature of humans and states), especially if the time frame is sufficiently long. Of course, our research goal is not to make everything endogenous but to identify the endogenous factors we wish to explain and the exogenous factors that help explain them. Our policy goal is to find factors that can be manipulated to change the outcomes. On an operational basis this means exogenous variables, but on a constitutional basis it means institutional parameters. The difference, of course, is that operational variables (e.g. monetary policy) may be effective even when changed fairly frequently (though this is debated) whereas institutional variables are valuable precisely because they change slowly.

The Number of Actors (n)

The determination of regime membership illustrates the tension between the exogenous and endogenous aspects of institutional scope. The point that scope both determines and is determined by its environment is fully apparent since actors and their characteristics appear here both as an independent and dependent variable.

The standard view is that collective action becomes more difficult as the number of actors (n) increases, but that this may be offset by increasing asymmetries among group members (Olson, 1965: 15; Oye, 1986).¹⁵ However, except for a few special cases such as the privileged public good group, there has been remarkably little detailed analysis of either the impact of changing numbers of actors or variations among their important characteristics. In addition, available evidence does not give as clear a conclusion as the Olsonian hypothesis would suggest. After briefly reviewing the differing conclusions from the CPR and IR literatures, this section examines the determinants of n and how institutions might affect those in desirable directions. The basic argument is that institutions play an important role in determining the number and character of participants in an issue and thereby mitigate the independent effect of n and actor heterogeneity (discussed in the next section) on institutional performance and cooperation.

Local CPR and IR literatures differ in their assessments of the importance of n . Ostrom (1990: 188–9, 212) concludes that n is overrated as a determinant of success or failure in collective action, so none of her design principles explicitly address it.¹⁶ She observes successful cooperation in very large groups, collective action failures in fairly small groups and no obviously strong correlation between group size and success. Yet Ostrom's diagnosis recognizes that cooperation is easier among small groups. She argues for the importance of 'nested enterprises' whereby individuals are organized through smaller groups that are then organized into larger groupings. Such hierarchical federation allows large group cooperation to be built upon the advantages of small group cooperation so that n is secondary to other considerations, especially organizational or institutional structure.

By contrast, the IR literature emphasizes n as a central factor for understanding international affairs. Adding or subtracting even a single state fundamentally changes an international interaction. Balance of power is the classic example. Despite wide disagreement over the precise relationship

15. Hardin (1982: 42–9) points out that n cannot change without some change in other elements of the problem, so this claim needs elaboration and modification according to the specific contextual circumstances. See also Bendor and Mookherjee (1987).

16. Two design principles, 'clear boundaries' and 'nested enterprises,' implicitly affect n as discussed below.

between numbers of major powers and the stability of the international system, there is broad consensus that even one great power more or less can make a decisive difference.¹⁷ A second example is that the zero-sum setting precludes cooperation among two actors but is fundamentally transformed once a third actor is included.¹⁸ The multilateral setting cannot be uniformly zero-sum and room for joint gain must exist at least among some of the possible coalitions. This has important substantive implications for international politics, including the circumscription of relative gains arguments as a basis for realist pessimism regarding cooperation among states (Grieco, 1990; Snidal, 1991). A final example is the contrast between games of collaboration and games of coordination which shows that the impact of increasing n is not uniform across cases. Increasing numbers impede cooperation in Prisoners' Dilemma games but not necessarily in coordination problems where increased n may increase the stability of cooperative outcomes (Martin, 1994; Snidal, 1985a). Thus while the direction of impact depends on additional factors, group size is important for collective action in IR.

In some local CPRs, the physical problem and available technology determine the number and nature of the actors. For historical or whatever reasons, a group of farmers who have fields along an irrigation canal (or fishers who live near a coastal fishing area) face a problem of joint provision or allocation of the common pool resource. The participants may be in a symmetric position and depend equally on the common pool or they may have asymmetric positions because some are located near the head of the irrigation system (or have historic priority to a particular fishing location). Regardless of the specifics, Ostrom (1990: 91) finds that clear 'boundaries' defining both the participants and the CPR are a key condition for successful resolution of CPR problems. Often these are set historically, sometimes independently of the CPR itself, and provide an exogenous basis for subsequent institutional development.

The historical evolution of local CPRs shows that boundaries are not always fixed and both the number and characteristics of participants fluctuate through time. Such change can destroy a successful CPR institution.

17. Gulick (1955) and Kaplan (1957) see the five-power system of 19th-century Europe as the paradigmatic situation for international stability, whereas Waltz (1979) believes bipolarity provides a magical recipe for stability. Recent more analytic treatments explore possibilities involving different numbers of powers (Niou et al., 1989; Wagner, 1986; see also Wagner, 1993).

18. Non-IR readers may feel I am stretching even the broad definition of collective action by including zero-sum settings where there are no collective gains to be made. However, the relative gains argument has gained prominence as a critique of the possibility of international cooperation because states maximize differences in payoffs (i.e. 'relative' gains, power). Moreover, important examples of collective action in international politics, such as alliances, clearly concern cooperation of one group at the expense of another.

An example is the Mawelle fishery in Sri Lanka where demographic and market pressures, combined with the inability of the local community to control its rules, resulted in the breakdown of a viable set of collective rules. Of chief importance was a rapid rise in the number of fishers combined with a shift from owner-producer to wage-labor production and a corresponding dilution of kinship groups. The success of the Törbel alpine pasture, by contrast, depended on a stable population maintained by late marriages, spaced births and emigration. The failure or success in each case depended on clear rules delimiting which actors had rights to the common pool. The Törbel case is particularly interesting because *n* was endogenous insofar as marriage and family plans were influenced by anticipated access to the CPR, while emigration provided the ultimate relief valve for commons access.¹⁹ In such situations, successful governance institutions limit the number of participants to what can be sustained within the productive capacity of the CPR.²⁰

The IR cases share these properties of both physical-technological and political determination of *n*. Geographic factors largely determine the necessary actors in some international environmental CPRs such as the Mediterranean basin. But in other CPRs, the participant states are fundamentally a product of international politics. Antarctica has been partitioned among a set of states with more or less arbitrary claims, just as the third world was a CPR (from a very ethnocentric Western view) partitioned among imperial rivals in the 19th century. Fishing grounds, navigation rights and seabed minerals all depend heavily on political agreements regarding property rights of individual states. The same is true for IR problems not analytically equivalent to CPRs. NATO mixes geographic imperatives with political considerations to define its problem as well as its membership. The GATT trading regime is not constructed around 'technological' considerations of comparative advantage that cut across North-South lines but, like the European Union (EU), has been primarily a club of industrialized countries. Technological considerations may even follow politics here in

19. The parallel in the Mawelle case was that CPR breakdown came at a time of changing connections to the external environment in terms of increased market pressures and improved transportation. This suggests that the self-regulating equilibrium of the Mawelle CPR was upset by outside intrusion on boundary rules.

20. Other examples in Ostrom et al. (1994) are also suggestive: Tang (1994: 322) notes that participation in a wide range of irrigation CPRs depends on landownership, irrigation-shares, membership in an organization or payment; Schlager (1994: 358, 368) notes that participation in a fisheries CPR depends on residency, use of certain technology, membership or a few other criteria; and Agrawal (1994: 375-6) notes that participation in Indian forest CPRs depends on residency, sometimes modified by contributions to maintaining the CPR. None of the cases offers a rich description of the origin or change of rules that could provide an interesting window into how endogenous institutions affect membership.

that, after an extended period of trading together, a set of countries may become inherently better suited to trading with each other (e.g. compatible standards, types and quality of goods produced) than with other countries. Thus, in both CPR and IR settings, n is defined by a combination of technological and political-institutional factors.

The most important way that political arrangements affect n is through mechanisms that exclude non-contributors or threaten punishment to induce contributions. Although pure CPR and public goods models assume exclusion is very difficult, significant exclusion occurs in many real settings. This is the result of conscious efforts (or of evolutionary good fortune) to overcome difficult collective problems by redefining them through political institutions. The CPR cases are instructive because they display how exclusion (or contributions) can be attained through a multiplicity of mechanisms even in situations where we should expect it to be most difficult.

The least relevant possibility from an IR perspective is exclusion or enforcement based on *external* political institutions. Ostrom's (1990) clearest examples are Los Angeles water basin problems where state courts determine the status and rights of various appropriators and thereby the basis for local bargaining towards an agreement. The failure of the Mawelle CPR previously mentioned rests in large part on the failure of external authorities to enforce boundary rules by preventing new entrants to the fishery. Finally, weekly water courts composed of representatives of adjacent Spanish irrigation districts (*huertas*) provide sanctioning power to each local CPR for the management of its irrigation agreements. Note that these CPR problems are not 'solved' externally, since local participants design the governing rules, but external enforcement of locally designed rules is what makes the solutions stick. Insofar as international anarchy is defined by the lack of such strong external institutions, however, international agreements must be enforced in different ways.²¹

At the other extreme, exclusion/enforcement can be *local and decentralized*. The best example is the 'Port Lameron', Nova Scotia fishers who defend their CPR territory through warnings and threats of destroying the equipment of intruders (Ostrom, 1990: 175). The assignment of fishing locations in the Turkish Alanya fishery is also enforced by the fishers themselves

21. Note that some forms of external enforcement operate at the international level. One is hegemonic enforcement where a dominant state enforces an agreement between two subordinate states, as the United States has in the Camp David accord. A more controversial example is whether the European Union is becoming sufficiently strong to represent a *de facto* external authority for some agreements between its members. Finally, an insufficiently understood case is where external authority is effective through the 'legitimation' of local rules rather than any direct enforcement apparatus. Ostrom (1990: 177–8) provides several examples where the opposite occurs and lack of external support undermines local rules.

since those assigned more desirable sites are prepared to defend them. Although such decentralized enforcement would seem to be a natural model for IR, the CPR examples indicate several drawbacks. An obvious one is the reliance on threats of violence by individual actors. Given the scale, potential and history of decentralized international violence, this is an unacceptably dangerous way to organize cooperation even if the evidence is that *use* of violence at the local level is infrequent.²² A second reason is that these solutions of CPRs are much more fragile than those rooted in external authority.²³ A final (and, in my mind, decisive) reason is that at least the Alanya enforcement regime is not completely decentralized, since enforcement of the rules receives support from other fishers who are not directly involved, but who value the maintenance of a rule system that improves their situation. The local coffeehouse emerges as a significant institution that provides a forum for conflicts to be resolved as well as a nexus for other social relations that support the fishing arrangement. Thus, even small-scale CPRs require more than strictly decentralized enforcement.

This leads us to the third mixed case where exclusion/enforcement of the CPR is managed locally but in a *locally centralized* way. The Alpine and Japanese mountain CPRs are essentially coextensive with the preexisting social and political fabric of their respective villages. The village assembly determines land rights, as well as membership and participation rights, both in the political community and in the commons. It provides a framework within which the CPR rules are decided, plays a direct or indirect role in the monitoring of those rules and provides the muscle behind the imposition of sanctions. This central connection is important even when Japanese detectives fine violators and take the proceeds for their personal use because it enables and legitimizes their seemingly decentralized practice. In addition, the informal networks and interconnections of small societies play an important role in supporting cooperative behavior since social ostracism is perhaps the greatest penalty. Thus, the commons are managed by a combination of formal and informal mechanisms that mimic some functions of the state at the local level. While there is no state in a strong sense, the setting for the resolution of these problems is not anarchic either.

IR resolutions of collective-action problems fall somewhere between these last two cases. Enforcement relies fundamentally on individual states but is facilitated and even orchestrated through more centralized efforts. As with the Japanese detectives, the rules of the international community justify

22. See note 5 above.

23. Port Lameron and Alanya are two of the three cases that Ostrom (1990: 180, Table 5.2) describes as fragile. The third is Gal Oya, Sri Lanka, which cannot be categorized in terms of this exclusion/enforcement trichotomy due to missing information on sanctions.

retaliatory behavior and distinguish such behavior from arbitrary violations of international norms. Moreover, even if they do not face direct retaliation, states worry about ostracism from international society just as villagers do from their society. It matters because their standing in the community facilitates their ability to achieve their other goals as, for that matter, does the maintenance of rules that promote collective ends.

The technology of many IR problems makes it easier for individual states and international institutions to determine n than in typical CPR cases. In particular, CPRs (and public goods) assume that exclusion is impossible for technical reasons whereas in many important international problems exclusion is a matter of choice. States decide with whom to trade and with whom to ally. When collective action is not tightly defined by physical imperatives, international communities may play a stronger role in defining their boundaries. In organizations like alliances and trading groups, benefits may accrue primarily to members or, even if there are spillovers to non-members, the additional benefits of membership may still exceed its costs. Moreover, there are often alternative ways to organize collective action (e.g. specific rules for a trading regime) with different distributional consequences for individual states. Now membership that provides a voice in shaping the content of collective action may be preferable to free riding.

Finally, local CPR cases raise interesting questions regarding the common presumption that the number of states involved in many international issues is a major impediment to cooperation. It is not uncommon for IR scholars to presume that small groups with fairly similar interests (e.g. the five states of G-5) will have difficulty cooperating and that cooperation among large numbers of states (e.g. G-77) must fail. A comparison to local commons problems suggests that IR theorists have overemphasized this problem. For example, the irrigation problems in the Spanish huertas involve upwards of 13,000 participants who are able to successfully resolve their collective action problem. Here the institutional arrangement of nesting (Ostrom, 1990: 184) mitigates the impact of large n . Even more telling is that 100 fishers – a number that would be large in IR – are able to cooperate in managing the Alanya fishery in a setting that is weakly institutionalized. Thus, international relations analysts must look beyond simple head counts to explain the failure of international collective-action problems since n and its impact are contingent on the institutional setting.

Heterogeneity Among Actors

Heterogeneity describes variation across actors on some significant attribute. Olson's (1965) famous privileged group argument is that an actor with a preponderant interest in a public good will unilaterally provide it to the

benefit of all. This has been imported into IR as the 'theory of hegemonic stability' (Kindleberger, 1973; Keohane, 1984) that the presence of a single dominant actor benefits the international system through its willingness to provide international public goods. Thus, the standard IR argument is that extreme asymmetry or heterogeneity promotes cooperation. By contrast, the CPR literature (Ostrom, 1990: 89, 210–13; Libecap, paper 7 in this collection) generally concludes that homogeneity promotes collective action. The striking difference in these claims arises because of their concern with different aspects of heterogeneity.

This section analyzes different types of heterogeneity and their impact on collective action in the two areas. The variety of forms and consequences of heterogeneity leads to a less than satisfying conclusion: the impact of heterogeneity is heterogeneous. Nevertheless, this is a useful advance over tendencies to conflate different types of heterogeneity and directs our attention to the differential impact of variations in actor characteristics. It thereby highlights key ways in which the design of institutions affects heterogeneity and thereby collective action.

Consider first the differences between CPR and IR arguments in terms of heterogeneity defined over capability. Hegemonic stability theory sees capability–heterogeneity as promoting cooperation because larger actors produce a public good whose benefits are joint across all states. Here benefits accrue to all, but costs fall on producers. By contrast, in CPRs the benefits of appropriation are not joint and accrue only to appropriators, while the costs of CPR depletion fall on all. These technological differences explain why capability–heterogeneity has a different impact across the two settings, since in the former case more powerful actors work in the common interest whereas in the latter case they work against it.²⁴

Arguments based on capability are typically inattentive to equally important variations in interests. In the hegemonic stability model, for example, interest and capacity are collapsed into willingness/ability to contribute to the collective effort.²⁵ This overlooks possible variations among states' interests by stipulating strong common interest in a public good. Now asymmetry (heterogeneity) is beneficial in the pure public goods model because

24. Moreover, homogeneity eases the tough bargaining problem in achieving cooperation on CPRs insofar as symmetry makes equal payoffs more compelling (Libecap, paper 7 in this collection). The same effect is likely with respect to a potential cooperating k -group in non-privileged public goods situations (Snidal, 1985b) – suggesting that the impact of heterogeneity is not monotonic for small groups.

25. The IR security literature distinguishes more clearly between interest and capability through concepts like resolve. In economics, by contrast, capability and interest are typically collapsed into the single concept of demand.

actors have different amounts of the *same* interest.²⁶ But when distributional issues are important because states have different interests – for example, if they have differences over *which* public good should be provided (e.g. debates over the purpose of NATO) – then heterogeneous preferences inhibit cooperation. In other situations, the impact of interest-heterogeneity can be completely reversed again. In market and issue linkage situations, for example, heterogeneous interests provide an essential basis for cooperative exchange whereas homogeneous actors have no incentives to cooperate (Martin, paper 4 in this collection). Thus, arguments about heterogeneity require careful specification since its different aspects need not have uniform impact.

Ostrom suggests that actors' time horizons and therefore discount rates are an important aspect of interest-heterogeneity and therefore determinant of collective action. The longer the horizon over which actors are concerned and the longer the CPR can be expected to operate, the more interest they will have in maintaining it. This reinforces the importance of institutions in lengthening participants' time horizons by creating boundaries that prevent non-contributors from appropriating from the CPR and, more generally, by increasing the stability of the CPR. However, heterogeneous discount rates, as well as heterogeneous preferences across goods at different points in time, can complicate collective-action problems. For example, industrialized and less-developed countries will find cooperation on the international environment difficult insofar as the latter are (quite reasonably) less willing to trade off current growth for longer-term environmental benefits. Thus, time horizons provide another aspect of heterogeneity that affects cooperation.

At least two additional types of heterogeneity are important for both IR and CPRs. One is that an issue may be populated by two or more fundamentally different types of actors. In this collection, Ronald Mitchell's discussion of maritime oil pollution (paper 9), for example, treats not only states but oil companies and independent shipping lines as significant actors while Kenneth Oye and James Maxwell (paper 8) focus on the pivotal role of individual private firms in the Montreal Protocol. Local CPRs also may be populated by differentiated actors, such as families or firms, or by fishers whose different technologies make their uses of the CPR fundamentally antagonistic. Many (but not all) of these differences can be captured by differences among capabilities or interests while others require a more serious consideration of subgroups as illustrated in the cited analyses. One of the interesting (but unexplored) institutional design questions at the international level is when, and in what way, states should allow non-state actors to participate in international rule-making.

26. Of course, public goods settings still produce mixed motives since even if actors all want more of the same public good, they still want others to pay for it.

A second important type of *non*-heterogeneity, or homogeneity, occurs when actors share vital characteristics that define them as a community. In discussing the virtues of homogeneity in resolving CPRs, Ostrom implicitly argues the advantage of common factors that tie the relevant community together and facilitate cooperation. The connections here may include common interest, but also go well beyond it. They are easiest to see in their absence (as in the Sri Lankan irrigation schemes that involve Sinhalese and Tamils from very different backgrounds) or when subgroups are in opposition to one another (as inshore fishers often are against offshore trawlers). Common language surely lowers the transaction costs of inter-Sinhalese (versus Tamil–Sinhalese) cooperation, but a common sense of identity and history also facilitates collective action within any homogeneous group. Similarly, inshore fishers have common interests, but it is probably more significant that they live together in small and fairly isolated villages and share no similar connections with the offshore trawlers. The same is true at the international level among states that share common cultural, geographic or historical experiences, whether in the Atlantic Community or the Islamic one.²⁷ Some of this effect is explained by the incentives offered by their multifarious interactions, but some of the effect goes beyond that to a shared sense of identity. The resulting community is supportive of collective action for the reasons discussed under ‘local and decentralized’ enforcement in the previous section. In any event, ‘community’ is a plausible candidate as a factor that reduces group heterogeneity in a way that promotes collective action.²⁸

This dimension of homogeneity raises the advantages and possible dangers of addressing international problems in terms of community subgroups. Subgroups can take advantage of their homogeneity, as well as their smaller size, to organize collective action. An optimistic view would see subsequent cooperation among these groupings paving the way to wider international

27. The danger of exaggerating this point is about as great as that of ignoring it. Japan has joined the ‘Western’ community because of its shared interests even though its historical-cultural legacy is very separate. The Islamic community too is clearly in the process of construction rather than being purely a predetermined affiliation. Oran Young (1994) provides a skeptical view of transferring arguments about the role of community in small-scale societies to the international level.

28. The partial success of the Gal Oya irrigation project in Sri Lanka, even though cultural heterogeneity was reinforced by the asymmetric locations of the respective ethnic communities in the irrigation system (with Sinhalese as ‘head-enders’ and Tamils as ‘tail-enders’), shows that lack of community is not fatal to collective action. Moreover, successful collective action provides a partial basis for developing community as suggested by the solidarity among the Gal Oya farmers during a subsequent period of ethnic violence (Ostrom, 1990: 170–1).

cooperation, as in Ostrom's conception of nested enterprises.²⁹ But several drawbacks to this form of solution are apparent. Obviously if exclusion is too expensive, then restricting participation to the subgroup is infeasible. Another problem is that subgroups may achieve only a fraction of the possible cooperative potential posed by the problem (Oye, 1986); ironically, the opposite danger is that they will achieve enough of the benefit to inhibit further extension of cooperation.³⁰ Current arguments about the dangers of regional trade hinge partly on this concern. Finally, increased subgroup *homogeneity* may increase the impact of heterogeneity in the larger group. The history of blocs in IR suggests the danger that subgroups can become antagonistic and create new problems much worse than the one being solved.

These several categories of heterogeneity can each be affected by political and social institutions. Boundary rules that affect n can also determine the composition of the membership. Criteria such as family and community membership in local CPRs, or sovereignty and liberal democracy in international politics, may create homogeneity that promotes collective action. In other cases, group boundaries might be chosen to increase heterogeneity because that increases joint benefits. Regardless, a key point is that heterogeneity is not strictly exogenous but rather is a consequence of the institutional arrangements through which collective action is organized.

Finally, are the actors in IR problems more or less heterogeneous than those in local CPR settings? The two settings seem similar insofar as the predominant actors, states and individuals respectively, are formally homogeneous in each sphere. But a brief look beneath this veneer reveals substantial differences among the actors in each setting. Individuals in local CPRs often vary in their interest in and capacity to affect collective action. Some will be located near the head of the irrigation system, others at the tail; some fishers have several nets, others only a share in one. Similar disparities exist in international politics. They are perhaps greater when comparing the very biggest and very smallest actors, or the richest and the poorest. But for many other groupings of states that confront collective-action problems, the intragroup disparities may be similar to those found in local CPRs. The one area where international politics is much less homogeneous than *certain* local CPRs is with respect to the 'community' conception of homogeneity. The isolated mountain and fishing communities almost certainly have deeper

29. This differs from the conception of 'minilateralism' (Kahler, 1992) where leadership by a few great powers produces multilateral collective action.

30. Technically, if subgroups are too effective there is no core and the collectively efficient outcome is not stable. This means a successful subgroup cannot expand its membership without either decreasing benefits of current members or treating new members less well. The resulting bargaining problems could prevent full collective action.

interconnections and stronger common identities than are found in international politics. But other local CPRs seem comparable to IR in the extent of 'community' and the overall importance of this form of homogeneity remains an open question.³¹

Conclusion: Lessons for IR from CPR

Institutional analysis concerns the three-way interaction between institutions, their environment and behavior. Although providing a relatively fixed *parametric* setting in which political behavior occurs, over the longer run institutions *endogenously* change in response to their environment and the choice of actors. The relationship is further complicated because the construction of institutions proceeds in part through the selection of the very environmental factors that are central determinants of the institutional form. Finally, these processes can occur in ways better described as 'evolutionary' and not directly the product of rational design, or in ways better described as 'rational' and thus primarily the result of conscious decisions, or most often as a complicated combination of the two.

CPR-IR comparisons focus research on the variety of cooperation problems and expose the dangers of over-reliance on particular models as paradigmatic in either area. There is important diversity not only across the two categories but also within each. Some differences are physical or technological as illustrated by the different levels of jointness or the technical difficulty of enforcing property rights. But political and social factors that determine the possibilities for achieving collective action are equally important sources of diversity. More importantly, the diversity is not always given but is subject to determination within an issue area. Thus, variables that have been taken as central exogenous determinants of international outcomes – such as the number of actors and their characteristics – themselves need to be explained. Other factors, such as the nature of anarchy, are seen not to be single-valued but rather to take on a range of different specific forms in particular circumstances. Finally, if international problems are messier and more difficult to resolve than local CPR problems, those differences demand explanation in terms of the politics and society of the two settings.

Comparisons across categories demonstrate the variety of solutions to collective-action problems and the accompanying variety of institutional roles. First, local CPRs show the importance of boundaries in defining cooperation problems and their possible solutions. International decision makers have long understood this (for better and worse) as reflected in

31. See Buzan (1993) for a recent discussion of international society.

decisions to exclude communist states from many postwar institutions, in the struggles by newly independent states to increase their influence in international organizations and in the multitude of G-groups constituted to address various international issues.

Second, the anarchy–hierarchy distinction that is often presented as the fundamental defining difference between domestic and international politics is grossly insufficient and misleading for both if taken too seriously. Broader forms of social relations among individuals that we characterize in terms of society or community provide alternative ways to organize cooperation that in certain respects are much more effective than coercive centralized institutions. While international society is less developed in some areas, in other areas the development of society compares favorably with that found in local CPR problems that have been successfully resolved.

Third, there is a wide range of institutional possibilities between anarchy and hierarchy and these intermediate forms may remedy the deficiencies of purely decentralized cooperation in ways that are more effective than is possible even through hypothetically strong centralized institutions. The expansive definition of international regimes including everything from norms to laws to expectations (Krasner, 1982) has been useful in identifying this range. But the IR field has been retarded in developing deeper understanding of the differential operation of these institutional forms and their relationship to specific functions of cooperation. Greater attention to informational considerations has been the best example of IR scholarship probing more deeply into the specific operations of international institutions that fall neither at the extreme of pure anarchy nor central government.

Finally, the local CPR literature provides a corrective to naive or overly optimistic views regarding the possibility of institutional design, as well as to unduly pessimistic ones. Successful CPR principles are not simply selected and then imposed on a situation. Ostrom's work is full of examples where well-intended central policymakers impose 'solutions' from above only to find their intentions frustrated by their lack of information regarding the underlying problem, by variations in the problem from one locale to another and, especially, by the unanticipated reactions of involved parties. Successful institutional design implicitly involves experimentation and learning as solutions are implemented and then modified in the face of results. The international politics corollary is that the absence of strong central authority is only a part of the problem. Even if we could establish world government – or a pocket of strong centralized authority over some aspect of international affairs – we could not solve such complex problems simply by writing rules and laws. Instead, we would need a process of learning and adapting rules to the problems and the states involved. Conversely, if we combine these 'evolutionary' lessons with what we can learn about rational design principles from local CPRs and other settings, then solving international collective-action problems need not rest heavily on strong central authority.

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DUNCAN SNIDAL is an Associate Professor in the Department of Political Science and the Harris Graduate School of Public Policy at the University of Chicago. His research focuses on issues of international cooperation and the role of international institutions. ADDRESS: Department of Political Science, University of Chicago, 5828 South University Avenue, Chicago, IL 60737, USA; snid@cicero.spc.uchicago.edu